ArogyaSaathi:Healthcare and Public Health Access

Voice-First Triage, Chronic Care Follow-up and Village-Level Outbreak Early Warning for Frontline Health Workers

Problem Statement

People in rural and remote areas face delayed diagnosis, missed follow-ups and late outbreak detection because doctors are far away and frontline health workers (ASHA, ANM, sub-centre staff) have little decision support. Patients with hypertension and diabetes are seen only at occasional appointments, so trends such as rising sugar combined with missed doses go unnoticed between visits. Health workers are unsure when to treat locally, refer or treat as an emergency. Signals of an outbreak in clinic visits, pharmacy sales and absenteeism are never combined, referral summaries are manual and inconsistent, and patients cannot describe symptoms or follow instructions in English or text-based systems. An AI that guesses, prescribes or misses a red-flag symptom can cause real harm, so strict guardrails, escalation and audit are essential.

Key Gaps Addressed

1) Missed deterioration between visits.

2) No decision support for local health workers on treat, refer or emergency.

3) Late outbreak detection from unlinked signals.

4) Language and literacy barriers for patients.

5) Safety risk from guessing, prescribing or missing red flags.

6) Poor continuity: manual, inconsistent doctor and referral summaries.

Objectives

#	Objective	Target
O1	Safe voice triage with danger-sign detection	100% of red-flag test cases escalated; 0 prescriptions generated
O2	Chronic care follow-up for BP and sugar	Deterioration flagged ≥ 7 days before next visit on simulated cohorts
O3	Decision support for health workers	Triage level agrees with clinician panel ≥ 90% (kappa ≥ 0.7)
O4	Community outbreak early warning	Detect injected outbreaks ≥ 3 days earlier than baseline; false alarms ≤ 1 per month per block
O5	Multilingual voice access	Symptom-extraction F1 ≥ 0.85 in 3+ languages
O6	Auditability and continuity	100% of decisions logged; doctor summary generated in ≤ 30 s

Users

Patients and families (voice-only, low literacy);

ASHA / ANM / community health workers (primary operators);

doctors at PHC or telemedicine hub (review escalations);

block and district health officers (receive outbreak alerts).

Exact Functionalities

- **Onboarding and consent:** registration by health worker or missed call; voice consent for data use, sharing and deletion.
- **Voice triage agent:** local-language speech-to-text, structured symptom extraction and guided questions using IMNCI-style and standard danger-sign protocols; output is Emergency / Refer / Treat locally / Home care.
- **Rule-based safety gate:** red-flag rules (e.g., chest pain, breathlessness, convulsions, severe dehydration) are deterministic and override the model; the system never prescribes or gives a diagnosis.
- **Chronic care agent:** scheduled voice check-ins for BP, sugar and medication adherence; trend analysis flags rising readings, missed doses and skipped visits.
- **Health worker decision support:** step-by-step checklist, treat-or-refer recommendation with protocol citation and uncertainty flag.

- **Referral and summary generator:** one-page structured summary (symptoms, vitals, history, actions) for the doctor, in the doctor's language.
- **Syndromic surveillance agent:** combines clinic visits, pharmacy sales and absenteeism; applies anomaly detection and spatio-temporal clustering to raise village or block alerts.
- **Doctor and officer dashboard:** escalation queue, patient trend charts, outbreak map and clinician feedback reused as labeled data.
- **Audit and oversight:** every input, rule fired, model output and human override is logged in a tamper-evident trail.

Backend and Database

Stack: FastAPI, LangGraph orchestration, Celery + Redis, Prefect ingestion, S3/MinIO for audio, Docker Compose.

Database: PostgreSQL with PostGIS (villages, facilities), TimescaleDB (vitals, visit counts, sales) and pgvector (protocol RAG).

Key tables: patients, consents, encounters, symptoms, vitals_series, medication_adherence, triage_decisions, protocol_chunks, red_flag_rules, referrals, surveillance_signals, outbreak_alerts, escalations, audit_log.

Security: pseudonymised patient IDs, encrypted PII, consent middleware, role-based and village-scoped access, DPDP-aligned erasure, prompt-injection defence.

Evaluation (No Field Trials)

Generate synthetic patients and outbreaks (e.g., Synthea-style records) with injected deterioration and outbreak patterns; clinician panel reviews triage on 200+ scripted cases; red-team tests for missed red flags, prescription requests and unsafe reassurance; backtest detection delay against baseline thresholds.

MVP Scope

Voice triage in 1 to 2 languages for 3 conditions (fever, child danger signs, chest pain), a hypertension and diabetes follow-up flow, rule-based red-flag escalation, a doctor summary, and one simulated outbreak alert on a village map.

Expected Impact

Earlier referrals, fewer missed deteriorations, safer health worker decisions and faster outbreak detection, with human oversight at every critical step.